

SMART CHILD SAFETY SYSTEM

A scalable African child protection ecosystem combining mobile technology, community intelligence, and wearable emergency response systems.

Every minute matters when a child goes missing. Yet across many African communities, emergency response systems remain slow, fragmented, or inaccessible during the most critical moments.

The Smart Child Safety System was designed to address this challenge by combining mobile technology, community intelligence, GPS/GSM wearable systems, and emergency response innovation to create a safer ecosystem for children across Africa.

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PROJECT OVERVIEW

The Smart Child Safety System is a two-version child protection ecosystem designed around two major objectives:

- Handle cases involving already missing children within the community (V1)
- Prevent child disappearances before they happen (V2)

Project Objectives

- Prevent child disappearances before they occur
- Accelerate the recovery process for missing children
- Create a real-time community emergency response network
- Use accessible technology adapted to African environments
- Improve emergency response efficiency and child protection

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SYSTEM STRUCTURE

The project is divided into two scalable versions designed to progressively build a complete child safety ecosystem.

V1 — Missing & Found Community Platform

Currently under implementation as the foundational layer of the ecosystem.

With respect to the higher technical complexity of V2, V1 is almost ready for implementation and deployment within approximately 2 weeks. This strategic approach allows the ecosystem to begin building credibility, user trust, and community adoption while the wearable infrastructure of V2 continues development.

V2 — Smart Wearable Protection System

A future expansion integrating discreet wearable GPS/GSM safety devices and intelligent SOS emergency systems.

Together, these two versions create a complete child safety ecosystem combining prevention, emergency response, and community collaboration.

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V1 — MISSING & FOUND PLATFORM

V1 focuses on managing cases involving children who are already missing through a mobile-based community platform.

Main Functionalities

- Report Missing Child
- Report Found Child
- Smart Matching System
- Community Alerts
- Real-Time Notifications
- User Profiles & Emergency Reports

The platform is designed to reduce response delays and improve coordination between families, communities, and nearby helpers.

REPORT AUTHENTICITY & VERIFICATION

To reduce false reports and improve trust within the platform, the system integrates an identity and child verification process before users can submit sensitive emergency reports.

Missing Child Reports

- Verify identity through secure account authentication
- Provide an official child-related document when available
- Support documentation may include birth certificates, school records, guardian proof, or medical identification

Found Child Reports

- Users must provide clear discovery information
- Community moderation and verification mechanisms validate reports before large-scale distribution

Benefits of Verification

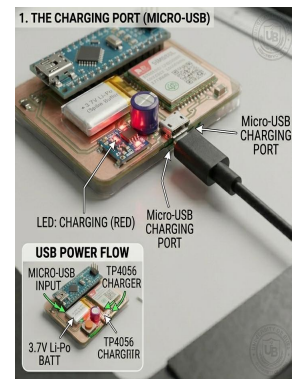
- Improve report credibility
- Reduce misinformation and fake alerts
- Increase trust between families and communities
- Strengthen child protection integrity within the ecosystem

The system is designed to balance accessibility, emergency speed, and information authenticity.

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V2 — SMART WEARABLE PROTECTION SYSTEM

V2 introduces a discreet wearable safety device designed to proactively protect children before a disappearance occurs.



Core Capabilities

- Track child movement in real time
- Trigger emergency SOS alerts instantly
- Send live alerts to parents and nearby community helpers
- Operate independently using GSM mobile networks

NB: In case some school discover the device and oppose them self to it, we endeavour to have a legal document from authorities that allow student to use the device in any area

WEARABLE TECHNOLOGY

SIM808 / SIM868 Module

Combines GPS, GSM, and GPRS communication in a compact module

GPS Technology

Provides accurate real-time location tracking

GSM/SIM Communication

Uses existing mobile networks to transmit alerts and emergency data

Physical SOS Button

Allows the child to manually request help during emergencies

Motion & Removal Sensors

Detect suspicious movement or forced wearable removal

Rechargeable Battery System

Sends a final emergency signal before shutdown or battery depletion

The wearable is designed to remain discreet, lightweight, and practical for everyday child usage.

SOS ALERT CONDITIONS

- Manual SOS button pressed by the child
- Child exits a predefined safe zone
- Wearable is forcibly removed
- Abnormal inactivity is detected
- Critically low battery level is detected

This multi-condition alert system improves reaction speed during dangerous situations.

PRIVACY & CHILD PROTECTION

- Parents do NOT continuously monitor the child's live location
- Real-time position is revealed only during emergency situations
- Secure backend infrastructure protects sensitive information
- Community access is monitored and controlled

This approach balances child safety with privacy protection.

HOW THE WEARABLE WORKS

1. The wearable collects location and sensor information
2. Data is transmitted using GSM/GPRS mobile networks
3. Backend servers process emergency events in real time
4. The mobile application instantly receives alerts from backend
5. Nearby community members can assist during emergencies

This creates a fast-response ecosystem capable of reducing emergency intervention delays.

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COMMUNITY NETWORK

Instead of relying only on parents or authorities, the platform introduces a community-assisted emergency response model.

ACTIVE MODE — "I CAN HELP"

- Receives nearby emergency alerts

- Can physically assist if available
- Helps reduce intervention time during emergencies

PASSIVE MODE

- Receives standard safety notifications only

The platform also includes a reward-based participation system designed to encourage positive community engagement and rapid response behavior.

This transforms nearby citizens into potential life-saving responders.

NB: The community network is not there to replace the police force but a way to fast reaction ,we endeavour to include the police force with time, just that we are limited to complexe procedures and modalities ,but for a fast lunch of the soluion we start with this approach.

TECH STACK

Frontend

- React Native
- Expo

Backend Infrastructure

- Supabase
- PostgreSQL Database
- Supabase Edge Functions

Authentication

- Supabase Auth

Notifications

- Firebase Cloud Messaging

Future V2 Technologies

- GPS/GSM/GPRS wearable communication
- Geofencing systems
- Real-time emergency tracking
- Embedded hardware systems

The architecture is designed to remain scalable, modular, and adaptable for future expansion.

EXPECTED IMPACT

- Reduce emergency response time
- Increase the recovery rate of missing children
- Build safer and more responsive communities
- Encourage community-driven child protection
- Provide scalable child safety technology for African societies

The long-term goal is to improve child safety accessibility through technology and collective action.

FINAL VISION

Our ambition is to build more than a mobile application.

- A scalable African child protection ecosystem
- A community-powered emergency response network
- An intelligent safety infrastructure capable of saving lives
- A sustainable technology solution for future generations

The Smart Child Safety System represents a vision where technology, community collaboration, and innovation work together to protect children more effectively across Africa.

TekMen Revolution never stands with folded arms while children continue to disappear without rapid response systems. That is why we transformed our vision into action — and today, we proudly present the first prototype of the Smart Child Safety System V1 on the link below.

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TEAM STRUCTURE

ZALI STEEVE : *PROJECT MANAGER AND FRONTEND DEVELOPPER*

TANTO EINSTEIN : *BACKEND DEVELOPPER*

TANYI FERDINAND : *HARDWARE ENGINEER*

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